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CLINICAL ARTICLE

Obstetrics

Effects of low-molecular-weight heparin and aspirin in recurrent pre-eclampsia: A stratified cohort study

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Abstract

Objective: To evaluate the effects of low-molecular-weight heparin (LMWH) combined with low-dose aspirin (LDA) in pregnant women with a history of pregnancy-related hypertensive disorders.

Methods: In the current retrospective stratified cohort study, 33 women with previous hypertensive disorders of pregnancy treated with LMWH and LDA were compared with 37 control women who did not undergo LMWH or LDA treatment. Rates of pre-eclampsia recurrence, placental abruption, and other adverse outcomes for the fetuses and pregnant women were compared in the two groups.

Results: The pre-eclampsia recurrence rates were 12/33 (36.4%) in the LMWH + LDA group and 28/37 (75.7%) in the control group ($P < 0.01$). In stratified cohort analysis, pregnant women with a history of early-onset pre-eclampsia, a body mass index of at least 24 (calculated as weight in kilograms divided by the square of height in meters), and aged less than 35 years benefited from LMWH + LDA treatment. In women with chronic hypertension or a history of placental abruption there was no protective effect. There were no significant differences in other adverse outcomes such as placental abruption and small size for gestational age in fetuses or pregnant women in the two groups.

Conclusion: Administration of LMWH + LDA only lowered the risk of pre-eclampsia recurrence in subgroups of pregnant women with a history of pregnancy-associated hypertensive disorders.

KEYWORDS
aspirin, hypertensive disorder of pregnancy, low-molecular-weight heparin, obstetric complication

1 | INTRODUCTION

Pre-eclampsia is a major obstetric complication that is often associated with premature delivery, placental abruption, and babies born small for gestational age (SGA). Patients with a history of pre-eclampsia are at an increased risk of pre-eclampsia in subsequent pregnancies. Women with previous severe pre-eclampsia reportedly have a 25%–65% risk of recurrent pre-eclampsia, a 3% risk of placental abruption, and a 10% risk of having an SGA baby in their next pregnancy.[1,2] Pregnant women with a history of pre-eclampsia have a five-fold greater risk of thrombophilia than those without a history of pre-eclampsia.[3] Facchinetti et al.[3] analyzed 172 cases of women with a history of pre-eclampsia in 2009, of which 34.9% were complicated with thrombophilia. Despite this, there are few reports of

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effective measures for the prevention of recurrent pre-eclampsia. It has been reported that therapy with low-molecular-weight heparin (LMWH) decreases the risk of placenta-mediated complications.[4] Notably, however, no studies have confirmed that LMWH combined with aspirin reduces the recurrence of hypertensive disorders and results in better pregnancy outcomes compared with either drug alone, in women with previous hypertensive disorders.

In the current study, whether LMWH combined with low-dose aspirin (LDA) could reduce the rates of pre-eclampsia recurrence and placenta-mediated complications in pregnant women with a history of pregnancy-related hypertensive disorders was investigated. Stratified cohort analysis was also conducted to investigate which subgroups of women derived the most benefit from LMWH + LDA therapy.

2 | MATERIALS AND METHODS

Patients were selected from the First Affiliated Hospital of Soochow University, Kunshan Friendship Hospital, and Kunshan Hospital of Traditional Chinese Medicine in China from December 2012 to May 2017. Women were eligible for inclusion in the study if they had a confirmed history of a diagnosed hypertensive disorder recorded in their previous obstetric records. Women were excluded if they met any of the exclusion criteria listed in Table 1. The study was approved by the Institutional Ethics Review Board of the Soochow University School of Medicine. Because it was a retrospective study and the names of patients were not divulged, informed consent was not obtained.

Seventy women were recruited, of whom 33 were given LMWH (2000–2500 IU) plus LDA (50–75 mg) daily, and 37 were not given LMWH or LDA. The 33 women in the LMWH + LDA group commenced the treatment at 12 weeks of pregnancy. LMWH was continued until 12 h before the onset of labor, and LDA was continued until 35 weeks of pregnancyor it was stopped if pre-eclampsia was diagnosed before 35 weeks of pregnancy.

TABLE 1 Inclusion criteria and exclusion criteria

Inclusion criteria
History of hypertensive disorders
Pregnancy confirmed by ultrasonography before 14 weeks of pregnancy
Randomization before 14 weeks of pregnancy
18 years of age or older
Exclusion criteria
Contraindication to heparin or aspirin therapy, including: thrombocytopenia, history of osteoporosis, active bleeding, documented peptic ulcer within 6 weeks, heparin or aspirin allergy
Severe damaged liver or kidney functions
Non-compliant patients (less likely to comply with necessary follow-up visits)
Previous participation in a heparin-related trial
Previous successful pregnancy with low-molecular-weight heparin or low-dose aspirin treatment
Requirement for low-molecular-weight heparin use such as previous thrombosis, antiphospholipid syndrome

All women received similar follow up during pregnancy and the postpartum period. Weight and blood pressure were measured and blood and urine were evaluated during prenatal check-ups. If the clinical manifestations and laboratory examinations were in accordance with pre-eclampsia diagnostic criteria, patients were advised to rest, and undergo spasmolysis and anti-hypertensive therapy. Recurrence rates of pregnancy-related hypertensive disorders were compared in the two groups. Occurrence rates of placental abruption, intrauterine death, SGA (birthweight below the 10th centile for gestational age), maternal complications (heart failure, acute renal failure, intracranial hemorrhage, HELLP syndrome [hemolysis, elevated liver enzymes, and low platelet count]), neonatal morbidity (neonatal asphyxia, respiratory distress syndrome), postpartum hemorrhage, full-term birth, gestational age at birth at least 34 weeks, and drug toxicity related to heparin (hemorrhage, skin reaction, thrombocytopenia [<100 000 cells/μl], bone fractures) were compared in the two groups. Mean birth weights were calculated in both groups.

Among the 70 women, indications for the termination of pregnancy included unmanageable hypertension, severe maternal complications such as HELLP syndrome, or the development of intrauterine fetal distress. The women were divided into subgroups based on confounding factors defined a priori, including body mass index (BMI, calculated as weight in kilograms divided by the square of height in meters), age, presence/absence of chronic hypertension, history of pregnancy, and severity of hypertensive disorders. Statistical analyses were performed with SPSS 20.0 (IBM Corp., Armonk, NY, USA). The Shapiro–Wilk normality test was used to determine whether continuous variables were normally distributed. Layered χ^2 tests or Fisher's exact tests were used to compare categorical variables, and t tests were used to compare continuous variables that were normally distributed. The non-parametric Mann–Whitney test was used to compare maternal age and BMI, which were not normally distributed. All statistical tests were two-sided and significance was set at $P < 0.05$.

3 | RESULTS

Maternal characteristics and pregnancy data in each group are shown in Table 2. The mean age of the women in the LMWH + LDA group was 31.6 ± 3.8 years (range 26.3–42.6 years), the mean BMI was 26.7 ± 3.8 (range 20–35), and 30/33 (90.9%) had a history of pre-eclampsia. The mean age of the women in the control group was 31.8 ± 3.0 years (range 26.4–40.3 years), the mean BMI was 25.0 ± 3.5 (range 15.6–32.8), and 23/37 (62.2%) had a history of pre-eclampsia. The pre-eclampsia occurred in 30/33 (90.9%) of the patients in the LMWH + LDA group and 22/37 (59.5%) in the control group (Table 2). There were no significant differences in maternal age, gravidity, parity, BMI, or underlying disease between the two groups (Table 2). The pre-eclampsia recurrence rate was significantly lower in the LMWH + LDA group than in

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TABLE 2 Maternal and pregnancy characteristics[a]

	LMWH + LDA (n = 33)	No treatment (n = 37)	*P* values
Maternal age, years	31.6 (26.3–42.6)	31.8 (26.4–40.3)	0.664
Gravidity			
<3	20 (60.6%)	18 (48.6%)	0.316
≥3	13 (39.4%)	19 (51.4%)	0.316
Parity			
<2	28 (84.8%)	1 (91.9%)	0.462
≥2	5 (15.2%)	3 (8.1%)	0.462
BMI	26.7 (20–35)	25.0 (15.6–32.8)	0.061
Pregnancy-labor history			
Pre-eclampsia	30 (90.9%)	22 (59.5%)	0.003
Gestational hypertension	3 (9.1%)	15 (40.5%)	0.003
Placental abruption	4 (12.1%)	2 (5.4%)	0.411
Intrauterine death	7 (21.2%)	5 (13.5%)	0.394
Underlying disease			
Kidney disease	2 (6.2%)	0	0.219
Cardiopathy	1 (3.0%)	0	0.471
Diabetes mellitus or gestational diabetes mellitus	2 (6.1%)	3 (8.1%)	1.000
Chronic hypertension	10 (30.3%)	16 (43.2%)	0.263
Others	1 (3.0%)	1 (2.7%)	1.000

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by the square of height in meters); CI, confidence interval; LAD, low-dose aspirin; LMWH, low-molecular-weight heparin.

[a]Values are expressed as median (25th and 75th percentile values) or number (percentage).

TABLE 3 Maternal and pregnancy outcomes[a]

	LMWH + LDA (n = 33)	No treatment (n = 37)	*P* values
Recurrent rate	12 (36.4%)	28 (75.7%)	0.007
Placental abruption	1 (3.0%)	3 (8.1%)	0.713
Intrauterine death	0 (0)	1 (2.7%)	0.769
SGA	2 (6.1%)	6 (16.2%)	0.131
Maternal complication[b]	1 (3.0%)	3 (8.1%)	0.869
Postpartum hemorrhage[c]	1 (3.0%)	0 (0)	0.813
Neonatal morbidity	2 (6.1%)	0 (0)	0.181
Full-term birth	18 (54.5%)	12 (32.4%)	0.078
Mean birth weight, g[d]	2673.94 (1080–3950)	2406.81 (1000–3950)	0.212
Drug toxicity	0 (0)	0 (0)	>0.99

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by the square of height in meters); LAD, low-dose aspirin; LMWH, low-molecular-weight heparin.

[a]Values are expressed as mean value (minimum–maximum) or number (percentage).

[b]Maternal complications include heart failure, acute renal failure, intracranial hemorrhage, and HELLP syndrome (hemolysis, elevated liver enzymes, and low platelet count).

[c]Postpartum hemorrhage is defined as a cumulative blood loss of greater than or equal to 1000 ml or blood loss accompanied by signs or symptoms of hypovolemia within 24 h after the birth process.[23]

[d]After exclusion of early and late loss (2 in LMWH + LDA group, 4 in control group) and fetal death (none); 33 neonates in the LMWH + LDA group and 33 neonates in the no treatment group.

the control group (36.4% vs. 75.7%, relative risk 0.48, 95% confidence interval 0.30–0.78, $P < 0.01$; Table 3).

In stratified cohort analysis of recurrence rate, patients with a pregnancy-associated history of early-onset pre-eclampsia had a lower post-treatment recurrence rate than those in the control group. Patients without chronic hypertension had a lower post-treatment recurrence rate than those in the control group. Patients without a history of placental abruption had a lower post-treatment

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recurrence rate than those in the control group, and patients whose BMI was ≥24 or were aged <35 years had a lower post-treatment recurrence rate than the control group (Table 4). Irrespective of whether patients in the LMWH + LDA group had experienced a previous intrauterine death or not, they had a lower post-treatment recurrence rate than patients in the control group (Table 4). There were no significant differences in mean birth weights between the LMWH + LDA group and the control group, and there were no significant differences in the occurrence rates of placental abruption, intrauterine death, SGA, maternal complications, neonatal morbidity, postpartum hemorrhage, full-term birth, or drug toxicity (Table 3).

4 | DISCUSSION

In general, prophylactic treatment with LDA and LMWH has protective effects with regard to reducing the recurrence rate of pre-eclampsia. Pre-eclampsia is characterized by an imbalance between two cyclooxygenase metabolites of arachidonic acid—thromboxane and prostacyclin—which may explain the main clinical symptoms of pre-eclampsia such as hypertension, platelet aggregation, and reduced uteroplacental blood flow.[5] LDA can selectively inhibit thromboxane synthesis, so it is considered a preventive therapy.[5] Heparin treatment is known to improve endothelial nitric oxide bioavailability.[6] LMWH causes dose-dependent relaxation in the human internal mammary artery via endothelium-dependent mechanisms, including the generation of nitric oxide.[7] As well as exerting anti-inflammatory effects and having beneficial effects on endothelial function, LMWH is considered a novel therapeutic option for preventing impaired placentation-related complications.[8] The precise mechanisms by which LDA and LMWH reduce the likelihood of pre-eclampsia recurrence are currently not well understood. The "FRUIT" randomized controlled trial conducted by de Vries et al.[9] indicates that adding LMWH to aspirin before 12 weeks of gestation reduces recurrent hypertensive disorder onset at <34 weeks of gestation in women with inheritable thrombophilia and a previous delivery with hypertensive disorders or SGA born at <34 weeks of gestation. It is plausible that prophylactic LMWH and LDA are beneficial for preventing recurrent pre-eclampsia in the thrombophilic subgroup of patients.

Notably, the use of LMWH and LDA benefit some subgroups of patients with a history of hypertensive disorders. Concordantly, in the current study, LMWH + LDA provided more benefit with respect to preventing pre-eclampsia recurrence in the early-onset pre-eclampsia subgroup than in the late-onset pre-eclampsia subgroup. Many researchers have suggested that early-onset pre-eclampsia is a placentally mediated disease, whereas late-onset pre-eclampsia is a disease of maternal origin.[10] Among the suggested hypotheses, the two-stage model[10] of disease is the most representative. In that model, the first stage involves a disorder of uterine spiral artery remodeling during placentation that leads to decreased placental blood perfusion. The second stage involves

TABLE 4 Stratified analysis of recurrent rate[a]

	LMWH + LDA (n = 33)	No treatment (n = 37)	*P* values	RR (95%CI)
Maternal age, years				
<35	8/27 (29.6%)	24/30 (80%)	0.001	0.105 (0.031–0.356)
≥35	4/6 (66.7%)	4/7 (57.1%)	>0.99	1.67 (0.496–2.744)
BMI				
<24	4/7 (57.1%)	13/14 (92.9%)	0.209	0.615 (0.319–1.188)
≥24	8/26 (30.8%)	15/23 (65.2%)	0.036	0.472 (0.246–0.903)
Pregnancy-labor history				
Gestational hypertension	1/3 (33.3%)	11/15 (73.3%)	0.245	0.455 (0.089–2.318)
Early-onset pre-eclampsia	8/25 (32.0%)	11/14 (78.6%)	0.008	0.407 (0.216–0.767)
Late-onset pre-eclampsia	3/5 (60%)	6/8 (75%)	>0.99	0.800 (0.352–1.816)
Chronic hypertension				
Yes	3/7 (42.9%)	12/16 (75%)	0.571	0.571 (0.232–1.407)
No	9/26 (34.6%)	16/21 (76.2%)	0.041	0.454 (0.254–0.811)
Previous history of placental abruption				
Yes	1/4 (25.0%)	2/2 (100.0%)	0.400	0.25 (0.004–1.365)
No	11/29 (37.9%)	26/35 (74.3%)	0.007	0.511 (0.308–0.846)
Previous history of intrauterine death				
Yes	2/7 (28.6%)	5/5 (100%)	0.028	0.286 (0.089–0.922)
No	10/26 (38.5%)	23/32 (71.9%)	0.018	0.535 (0.314–0.911)

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by the square of height in meters); CI, confidence interval; LAD, low-dose aspirin; LMWH, low-molecular-weight heparin.

[a]Values are expressed as number (percentage).

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clinical terminal organ damage as the result of the release of active substances from the intervillous space into the maternal circulation, which exacerbates a systemic inflammatory response in maternal blood vessels resulting in reduced effective blood volume and enhanced vascular reactivity. The second stage may be common to both early-onset and late-onset pre-eclampsia, whereas vascular remodeling disorder in the first stage is more obvious in early-onset pre-eclampsia.[10,11] Crispi et al.[12] reported that placental growth factor was lower and sFlt-1 and vascular cell adhesion molecule-1 were higher in mothers with early-onset pre-eclampsia than in controls. LMWH is anti-inflammatory and has beneficial effects on endothelial function,[13] therefore LMWH + LDA may provide greater benefit with regard to preventing recurrent pre-eclampsia in women with early-onset pre-eclampsia than in women with late-onset pre-eclampsia.

In the current study, LMWH + LDA treatment bestowed an advantage on women who did not have chronic hypertension. Chronic hypertension is a systemic vascular disease, the main manifestation of which is small artery spasm in the early stages. Increased disease severity leads to arterial wall thickening, luminal stenosis, and arteriosclerosis, all of which substantially affect circulatory system function.[13] Chronic hypertension is a known risk factor for hypertensive disorders,[14] and clinical studies indicate that 100 mg LDA daily has an antihypertensive effect in untreated patients with mild chronic hypertension when taken before bedtime, but it has no antihypertensive effect when taken in the morning.[15,16] With regard to mechanism, it may be that aspirin is a potent antioxidant that can reduce the production of oxides in blood vessels, reduce nitric oxide release, and prevent angiotensin II-induced vasoconstriction.[15,16] Hence, aspirin may benefit treatment-naive women with mild chronic hypertension. Aspirin and LMWH may not have a satisfactory preventive effect in women with a long history of chronic hypertension.

In the present study, LMWH + LDA had a greater preventive effect in women whose BMI was 24 or more compared with women whose BMI was less than 24. Results of previous controlled cohort studies suggest that women with a high pre-pregnancy BMI are at greater risk of pre-eclampsia.[14] High BMI is strongly associated with insulin resistance, type 2 diabetes mellitus, and gestational diabetes mellitus. Obesity and diabetes mellitus are significant risk factors for cardiovascular diseases, and can aggravate the symptoms of pre-eclampsia,[17,18] and insulin resistance and hyperglycemia can impair endothelial cell function.[13] Prophylactic use of LMWH and LDA may have stronger effects in women with higher BMIs before pregnancy.

In patients who had a history of placental abruption in the current study, recurrent pre-eclampsia occurred in 1/4 (25%) of those treated with LMWH + LDA and in 2/2 (100%) of those who were not (P = 0.400). In patients who had a history of pregnancy resulting in intrauterine death, recurrent pre-eclampsia occurred in 2/7 (28.6%) women treated with LMWH + LDA and in 5/5 (100%) who were not (P = 0.028). This suggests that LMWH + LDA can decrease the likelihood of pre-eclampsia recurrence in women with a history of adverse events during pregnancy such as placental abruption and intrauterine death. In a previous study, in a small subgroup of patients with previous abruption LMWH may have prevented placenta-mediated complications in subsequent pregnancies, but strong evidence is lacking and multicenter trials are required.[19] Placental abruption is considered a bleeding disease, so it is important to exercise caution when considering administering LMWH to women with a history of it.[19] Intrauterine death may reflect the seriousness of hypertensive disorders. LMWH + LDA treatment can decrease the recurrence rate of pre-eclampsia and improve pregnancy outcomes.[20]

Administration of LDA and LMWH does not decrease the risk of placental abruption, intrauterine death, or SGA. Cumulative observational scientific literature suggests weak associations between thrombophilia and placenta-mediated pregnancy complications including severe pre-eclampsia, SGA, and abruption.[21,22] There is no definitive evidence that placental thrombosis is a major contributor to placenta-mediated pregnancy complications. Placental abruption is a bleeding complication. LMWH may prevent placental infarction that often precedes bleeding into placental infarcts, which manifests clinically as placental abruption.[19] Intrauterine death can be caused by umbilical cord factors, amniotic fluid factors, placental abruption, and fetal malformation. LMWH and LDA may have a therapeutic effect by way of prevention of intrauterine death caused by placental thrombosis.

Treatment with LDA and LMWH does not increase the risk of postpartum hemorrhage. A meta-analysis from 2014 indicates that LDA does not increase the risk of postpartum hemorrhage and intracranial hemorrhage, and it does not increase perinatal mortality.[23] Although data are lacking on long-term prognosis, LDA is considered safe and therefore does not necessitate additional monitoring. The results of randomized controlled trials on withdrawal times differ, but most suggest withdrawal of LDA at the time of delivery, and some studies suggest withdrawal of LDA at 35 weeks of pregnancy (before delivery), or when pre-eclampsia occurs.[24] Compared with ordinary heparin, LMWH has a lower capacity for binding to plasma protein, better bioavailability after subcutaneous injection (90%), and a half-life that is approximately two to four times that of ordinary heparin (2.8–4.1 h vs. 1.0 h). The anticoagulant effect is clearly dose-dependent. The antithrombotic effect is enhanced two- to four-fold, and adverse effects such as bleeding are lower in patients treated once or twice with LMWH. The therapeutic effect and safety can be assessed by monitoring coagulation factors and platelet count.

In general, treatment with LDA and LMWH has a protective effect by way of reducing the likelihood of pre-eclampsia recurrence. In stratified cohort analysis pregnant women with a history of early-onset pre-eclampsia and a BMI of 24 or more benefited from treatment. In women with chronic hypertension or a history of placental abruption or intrauterine death there was no protective effect. There were no significant differences in other adverse outcomes such as placental abruption or SGA between women who took LDA and LMWH and those who did not.

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CONFLICTS OF INTEREST

The authors declare that there are no conflicts of interest with regard to the publication of this article.

AUTHOR CONTRIBUTIONS

YC, WY, and FS contributed to the study design, literature search, and data acquisition and interpretation. YZ and FS performed the major data analyses, statistical analysis, and manuscript editing. JW and JZ contributed to data acquisition, statistical analysis, and manuscript revision.

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